

Nikolay Malkovsky

Curriculum Vitae

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Work Experience

- 2023–Present **Principal Researcher**, *Huawei*, 2012 Laboratories
- Improved the QUIC protocol for file-transfer and real-time communication scenarios using congestion-control algorithms
 - Technical lead of an R&D project on forward error correction and its applications to real-time data transfer, including advanced Reed–Solomon and convolutional-code implementations
 - Author of the TechArena 2025 SPbSU challenge
- 2022–2023 **Developer**, *Yandex*, Yandex Market
- 2021–2022 **Senior Researcher**, TCS Group, AI Department
- Transformer-based zero-shot text cleaning
 - Developed core components for CTC-based end-to-end ASR systems and WFST-based context biasing
- 2018–2021 **Researcher/Senior Researcher**, STC Innovations, Research Department
- Developed the HMM component of hybrid HMM-DNN ASR systems, including finite-automata-based solutions for word-boundary extraction, low-latency streaming recognition, vocabulary expansion, lattice generation, conversion to confusion networks, and lattice rescoring with RNN language models
- 2017–2022 **Assistant Professor**, St. Petersburg State University and the St. Petersburg campus of the Higher School of Economics
- Taught a course on convex optimization in Russian

Open Source

- 2025–Present **Author and Maintainer**, *Pixie on GitHub*, C++20 succinct data structures library
- Developed highly optimized compressed data structures that provide fast access without full decompression; focused on practical memory efficiency and CPU performance

Languages

- Russian: native; English: fluent

Education

- 2013–2017 **Ph.D. in Computer Science (Candidate of Science)**, St. Petersburg State University, St. Petersburg, Russia
- **Ph.D. thesis:** “*Randomized resource distribution algorithms in multiagent systems*,” supervised by **Prof. Oleg Granichin**
- 2008–2013 **Master’s Degree**, GPA: 4.5/5, St. Petersburg State University, St. Petersburg, Russia

Skills

- **Strong algorithmic skills**
Developed performance-critical real-time systems across several application areas, including graph and finite-automata algorithms for ASR, finite-field computations and coding theory for data protection, and state-of-the-art succinct data structures
- **Strong with C/C++**
Primary programming languages; led and maintained several algorithmically rich C/C++ projects of medium size (10^5 – 10^6 lines of code). Experienced in hardware-aware optimization, including cache-locality improvements and manual SIMD implementations.
- **Strong mathematical background**
Candidate of Science in computer science (Candidate of Physical and Mathematical Sciences in the Russian degree system). All of my major projects have involved substantial mathematics.
- **Strong with Python**
Primary language for scripting and prototyping, typically using Jupyter, NumPy, SciPy, scikit-learn, Matplotlib, and PyTorch/TorchScript. For performance-critical bottlenecks, I develop C/C++ libraries and invoke them from Python.
- **Strong with ASR frameworks**
Developed projects based on Kaldi, DeepSpeech, and NVIDIA NeMo.
- **Microservices**
Experience building microservices with Docker, gRPC, and Kubernetes.
- **Technical leadership**
Led a group of algorithmically skilled undergraduates, including two first-year students with IOI gold medals. Established clear workflows that enabled them to contribute high-quality code despite limited development experience.